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Chatbot personality preferences in Global South urban English speakers

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ABSTRACT

Recent years have seen a major shift in focus when it comes to chatbot literature, moving from only considering their technical capacities to emphasizing their social and conversational abilities, as well as personality. This shift towards the more social end of the AI research spectrum possibly occurred as a response to social interface theory, which suggests that humans respond to technologically mediated communication in the same way as they respond to interactions with natural partners (i.e. other human beings), since both interactions are social in nature. Given this, many prominent personalities in the sphere of chatbots and artificial intelligence have wondered if personality may play as important of a role in HCI as it does in social relations. To that effect, two studies were carried out in the Global South, on a largely South Asian demographic to see if personality does influence user experiences with reference to enterprise chatbots across three domains: likelihood of re-engagement, comfort levels and estimation of productivity and efficiency. Three key personality types were developed using the Five Factor Model and IBM's Primary and Secondary Characteristics Table, labelled Transactional, Prosocial and Friendly. The first study involved having participants interact with the three chatbot personalities and then rate them across the aforementioned domains, while the second study had participants view interactions with the different chatbot personality and provide reviews. The results strongly suggest that inserting an evident, fun personality into the chatbot can influence user experience across these domains.

1. Introduction

In psychology, personality is often defined as 'a pattern of relatively permanent traits and unique characteristics which lend an individual's behaviour some amount of consistency and individuality' (Feist & Feist, 2009). While a multitude of other definitions exist, what this definition innately suggests is that personality lends stability to human behaviour, with stability acting as a measure of how likely an individual is to exhibit a certain behaviour across time and situations (Carver & Scheier, 2012). This sense of continuity allows personality to provide us with a framework to explain the roots and causes of human behaviour and often causes personality to be considered as a 'fundamental linchpin in human relationships' (Dryer, 1999). Various studies have shown that personality, in the context of human interaction, allows us to classify the behaviour of our social partners into simplifying traits, predict other people's behaviour and guide our own responses to others, suggesting that it plays a vital role in the development of social relationships (Schultz & Schultz, 1998, p. 348; Uleman et al., 1966). Recent research into personality compatibility also suggests that personality can affect how satisfying a social relationship can be, with an interaction with someone who has a compatible personality can make the overall

experience far more productive and enjoyable for all those involved (Horowitz et al., 1977). In fact, it can be argued that in psychology, personality is often considered key in maintaining and establishing social relationships.

The study of personality, however, is by no means a new area of research, with interest in human personality going back to the time of Hippocrates and his theory of the Four Humors. This particular theory suggested that there were four basic bodily fluids, whose correct proportions defined 'health' and relative balance affected the human temperament and manifestations of behaviour — essentially referring to personality (Kalachanis & Michailidis, 2015). This theory was further developed by Galen, who created a typology of temperaments that were defined by the presence of these substances in the human body, in the belief that they could influence human emotions, behaviours and moods (Flaskerud, 2012). While no longer considered a part of mainstream literature in personality psychology, these theories formed the foundation for existing literature and influenced the development of a holistic view of personality.

Although the study of personality today is inundated with various models and theories, the model that perhaps enjoys the most fame is the Five Factor Model, often abbreviated as FFM (Costa and McCrae, 1987).

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Also known as the Big Five model, it consists of five global factors or traits that were found as a result of lexical analysis rather than any neuropsychological experiments (Goldberg, 1993). These five factors were the most common descriptive terms used to describe other people and represent a range between the two extremes of the personality traits. These traits are as follows:

- ❶ Openness to experience, people who score high on this trait often are more willing to try new things and are amenable to new thoughts and beliefs that challenge their existing assumptions.
- ❷ Conscientiousness, people who score high on this trait tend to be responsible and prepared, as well as more aware of the consequences or results of their actions than others
- ❸ Extraversion, those who are high on extraversion tend to be more outgoing and sociable than those who score lower
- ❹ Agreeableness, people high on agreeableness generally tend to be altruistic, likeable and cooperative
- ❺ Neuroticism, people scoring high on neuroticism are often chronic worriers, and get tense and anxious in stressful situations

Various empirical studies support the universality of the FFM across languages, ages and cultures, making it a fairly standard model for the study of personality across various domains, including workplace relationships, marital happiness and peer relations (Gurven et al., 2013). In fact, efforts have been made to apply the various factors to develop a framework of animal personality as well, with research suggesting that the dimensions of Extraversion, Agreeableness and Neuroticism show cross-species generalizability (Gosling & John, 1999).

Although the study of personality has been applied far and wide, and across several domains and industries, it was limited largely to the study of sentient individuals, by virtue of the belief that they were the only ones who could develop a personality — in fact, research on animal personality is still a fairly nascent field (Maestripieri, 2009). However, the explosive rise in the use of chatbots (short for chatterbots) and voice assistants by companies in the last decade as a means of automating lower-level customer service tasks, such as FAQs and delivery tracking, has caused personality to become an attribute that even artificially intelligent conversational agents must exhibit, albeit through text and audio instead of the more traditional modalities of speech and body language (Chakrabarti, 2012; Crutzen, 2011).

While much of the previous focus in chatbot and artificial intelligence development had been on their functional capacities, the need to create chatbots that could help in customer retention and engage with consumers on a more personal level resulted in an increasing movement towards enhancing their social aptitude (Klooster, 2017). Certain chatbots, such as Microsoft's Xiaoice or Steve Worswick's Mitsuku (four-time winner of the Loebner Prize) that are designed expressly for the purpose of casual chit-chat are already frontrunners in the race to give A.I. personality, displaying the personalities of eighteen-year-old high school girls. They often engage in casual flirtation with users, attenuate their responses based on positive or negative cues, have their own opinions and can converse about a variety of topics, often peppering their messages with wry remarks, conversational detours or jokes that mimic actual human interactions (Zolfaghari, 2015).

Xiaoice, in particular, seems to be the Holy Grail of personality research in artificial intelligence so far, using an Empathic Computing Framework and deep learning techniques to absorb skills from the interactions it has with its users — slowly acquiring their behaviour and social savoir-faire over time, until it seems to have developed a persona of its own (Spencer, 2018). Xiaoice's design architecture focuses as much on developing the EQ of the chatbot as it does on the range of skills that it possesses, incorporating empathy and social skills into the development of the chatbot and endowing it with skills like user profiling, emotion detection and sentiment recognition in order to develop responses that are situationally appropriate, motivating and positive for the user (Zhou et al., 2018).

Even transactional or service oriented chatbots, traditionally limited to only a few operations, have started developing their own personalities in a bid to enhance user experiences. For example, Cleo, a financial services bot deployed on Facebook Messenger displays a sassy, tongue-in-cheek personality when chatting with her mostly millennial users about their spending; often using humor and cracking sly jokes about their spending in order to keep them on track. In a bid to further attract their target demographic, Cleo recently introduced a 'Roast Me' feature that involved her dishing out some 'tough love' to users about their finances (MeetCleo, 2019). The feature proved to be immensely popular, with users taking to Instagram, Twitter and Facebook in droves to share their exchanges with Cleo, suggesting that personality does add some sort of draw to an otherwise fairly mundane interaction.

The recent emphasis in chatbot literature on improving their social and conversational capacities, alongside personality development possibly occurred as a response to the Media Equation Theory (Jain et al., 2018b; Reeves & Nass, 1996), which suggests that people respond to computer mediated interactions in the same way they respond to actual human interactions. Since the interactions people carry out using technological means are also fundamentally social in nature, it triggers an automatic reaction that leads people to unconsciously attribute different personality characteristics to the computer, depending on the interaction or even get frustrated when it fails to live up to their expectations (Lee, 2008; Reeves & Nass, 1996). In fact, studies suggest that media equation occurs because the human brain has not evolved to deal with twenty-first century technology and may be unable to distinguish between the real world and the virtual — an idea supported by the fact that the brain processes emoticons and human faces at similar occipitotemporal sites (Churches et al., 2014).

The idea that humans interact with computers in much the same way that they interact with other humans holds several key implications for the field of HCI [human computer interaction]. If personality compatibility is a key instrument in ensuring relationship satisfaction and interpersonal closeness between humans, and humans react to machine-mediated and human-mediated communication in analogous ways, then it is possible that personality can be leveraged in order to inspire the same effect in human-computer interaction as it does in human-human interaction (Berry et al., 2000; Long & Martin, 2000, pp. P311–P319). Research suggests that software agents, and chatbots in particular, have certain qualities that result in an increased likelihood of eliciting social responses from users, such as their ability to use full-text sentences alongside other typical forms of communication like icons or menus (Dryer, 1999). Full text sentences create a more natural experience, especially for people who possess a verbal cognitive style while the chatbot's ability to guide users to perform tasks or provide them with information gives it a sense of intelligence (Horn & Cattell, 1966). Similarly, another feature that helps evoke social responses from users is the agent's ability to execute actions on the behalf of the user, giving them a degree of autonomy that other technical systems may not possess (Reeves & Nass, 1996).

HCI interactions that can induce the same type and amount of sentiment as human-human interactions hold a lot of appeal for the field as a whole, but are especially interesting for the nascent industry of enterprise chatbots, leading to ongoing research into what sort of personalities users would like to see in the chatbots that they are increasingly beginning to communicate with. An exploratory Wizard-of-Oz [WoZ] study conducted by Microsoft researchers focused on understanding what sort of personality was preferred by young, urban users (Thies et al., 2017). The participants in the study interacted with a Wizard, a human operator who simulated three separate, pre-defined personalities over Skype Messenger in order to carry out the study in the absence of the actual chatbots.

The study found that urban users had a tendency to lean towards the simulated chatbot personality that was productive yet friendly, and thereby able to add some sort of value to their life during the interaction. In comparison, while they did enjoy interacting with the flirtatious

personality, they soon grew tired of her high energy while the ‘listening bot’, who acted like a shoulder to lean on was widely regarded as overly caring and not very interesting to talk to, although some participants did appreciate her listening ability. Another takeaway from the study with regards to personality included the discovery that users tended to look for humor when interacting with chatbots, something that had also been previously confirmed by various other studies (Morkes, Kernal & Nass, 1999). Since humor is often considered to be indicative of high psychometric and emotional intelligence, embedding chatbots with humor may result in an increased perception of them as autonomously intelligent and add to their personality, acting as a further draw to users and helping reduce any cognitive dissonance that the interaction may result in (Cooper et al., 1978; Hauck & Thomas, 1972). Given the speed at which the Global South, and specifically South Asia is beginning to adopt these technologies and use them in daily life, as well as the dearth of literature on this topic, this study aims to bridge this gap in literature by attempting to replicate and validate the results of the aforementioned study using real chatbots rather than a simulated WoZ protocol in order to understand if personality can affect user preferences and perceptions of chatbots, and therefore influence how likely the users would be to re-interact with the chatbot.

2. Experimental overview

In order to test whether personality has an influence on user preferences and perceptions of chatbots, this study used a mixed method approach that attempted to gather both quantitative and qualitative data. Prior to describing the experimental conditions, the study will first describe in detail the development of the chatbots, including the focus group used in its development. Following this, the paper will describe Experiment 1, which used a food-ordering test case to examine the roles of three types of personality on users by rating them using Likert scales and then carrying out a statistical analysis to find underlying trends in the data. Subsequently, the paper will describe Experiment 2 and the use of a semantic thematic analysis to understand the qualitative data underpinning human-chatbot interactions in a different test-case with a different demographic.

Experiment II can be considered a conceptual replication of Experiment I and attempted to understand if the preference for chatbots with evident personalities was generalizable across use-cases and generations. Instead of using a demo food-ordering bot, these bots tested user reactions to an online chat-based dictionary that was available in three different chatbot personalities. However, rather than interacting directly with the chatbots themselves, participants were asked to view.mp4 files of pre-scripted interactions on a Facebook Messenger interface that was developed using Botsociety, a prototyping conversational interface instead of Haptik.Ai’s Bot Builder Tool. Following this, participants were then asked to give their opinion on which chatbot they would like to interact with and why. Two key differences across the studies was the use-case that was being tested, with Experiment I testing a productivity-based use-case while Experiment II focused on an educational use-case. This was done in order to see if the preference for a friendly personality would be generalizable across use-cases and not only relevant to one. Secondly, there was a marked age difference in the age and range of the participants across the two studies. Although this was not necessary, the experimenter believed that it would be pertinent to test whether the want of a friendly personality would be similar for current users (participants in Experiment I) and future users (participants in Experiment II), since it may be possible that future users would react differently to the personalities given the fact that they are far more ‘wired in’ and aware of technological developments than their predecessors.

Informed consent was taken from all participants, and permission taken from parents of participants who were under the age of 18. Participants were also made aware of their right to withdraw and the privacy of their data.

2.1. Theoretical rationale behind chatbot development

Various studies around social interface theory have already supported the idea that humans react to technologically mediated partners like software agents in the same way as they respond to natural partners, often transferring their knowledge of social and natural experiences to help make sense of their interaction with a technologically mediated partner (for a greater overview, see Reeves & Nass, 1996). Given this, it is fair to assume that personality could act as a linchpin for not just social relationships with other humans, but with user retention for bots as well (Dryer, 1999). Three key personality types were tested in this study, based on the most wide-spread type of personalities that tend to appear in chatbots — transactional, prosocial and friendly (Baker, 2019).

Much like other personality based A.I. research, this study also chose to adopt the FFM in order to define the personality traits. However, since the bots were not built using a generative model, they could not generate a personality by themselves based on a corpus database. Instead, any semblance of personality was manually added in the copy rather than being generated automatically. Since the copy was written on the basis of the experimenter’s idea of how each personality type should be expressed, certain measures were taken to ensure that they were appropriately demonstrating the various personality dimensions. In order to bolster the experimenter’s representation of each personality type and the associated dimensions, IBM’s Primary and Secondary Characteristics Table, which is a part of their Personality Insights module, was consulted prior to the writing of the copy (IBM Cloud, 2019). The chatbots were built using pre-chosen descriptions of three key traits taken from the table: Conscientiousness, Agreeableness and Extraversion, in three variations that were believed to best represent the three personality types that the experimenter was attempting to study. Following this, sections of text from the interactions were processed via IBM’s Emotional Analyser in order to see whether the chatbots were appropriately demonstrating the personality assigned to them while interacting with the user.

2.2. Detailed Chatbot Personality Outline

A detailed description of each of the three chatbot personality is delineated below:

Chatbot A: Has an extremely transactional personality, reflected in both in its name “WcBot” and tagline, which reads as “AI Powered WcDonald’s Chatbot”. Efficient and focused on carrying out tasks with the lowest numbers of interactions possibly, the personality of this chatbot can be defined as highly principled and serious. It does not interact socially with users apart from basic niceties that help move the conversation forward. The copy for this bot was extremely basic, using only short, simple words and avoiding any indicators of engagement, such as emoticons or exclamation marks. On the basis of IBM’s Characteristics table, this chatbot’s primary trait was Conscientiousness, under which it had a high score but scored low on its secondary trait of Extraversion, defining its personality as restrained, serious, disciplined and cautious. This was supported by the results of the Emotion Analyser, which rated the dominant tone in the text as ‘analytical’ (0.60), which is described on the website as text that is likely to be perceived as ‘rational, systematic and impersonal’ (See Fig. 1, leftmost chatbot).

Chatbot B: Unlike the WcBot Chatbot, this chatbot was designed to display a more prosocial personality. In order to further advance this impression, steps were taken to create a more anthropomorphic personality. Instead of being referred to as a chatbot, this particular demobot was given the human name of “Ronald WcDonald”, as well as the tagline “Here to help you order!”, which aimed to be reminiscent of customer service interactions. This was done in a bid to understand if making the chatbot personality somewhat anthropomorphic would increase comfort levels in the users, although it should be noted that in all the use-cases, the bot introduced itself as an artificially intelligent

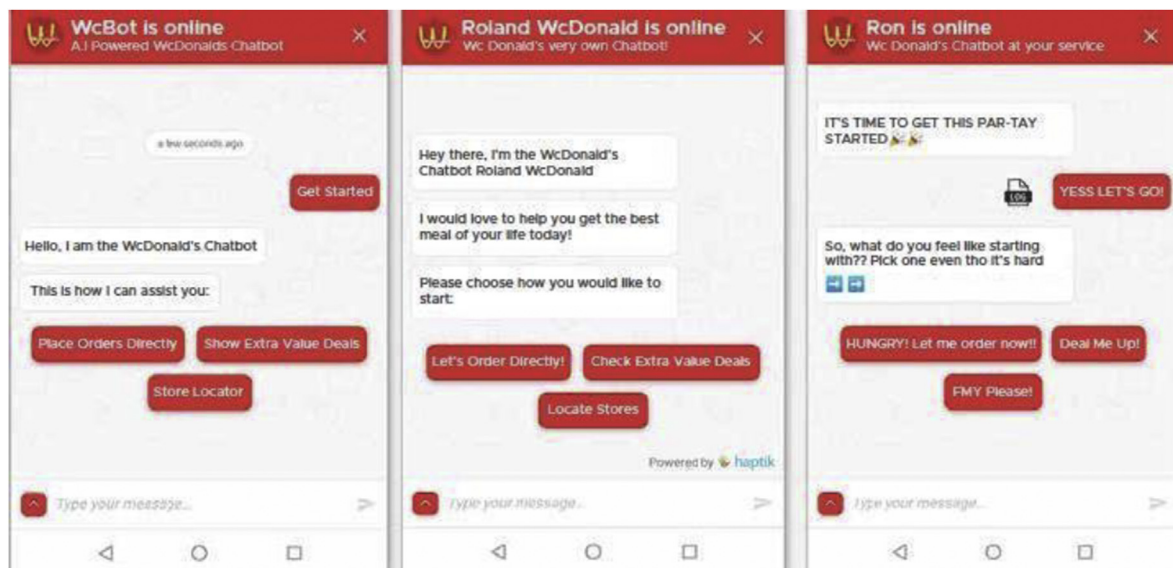


Fig. 1. Transactional, prosocial and friendly food ordering chatbot.

conversational agent in order to avoid any form of deception. The personality of this bot could be described as mellow and helpful, aiming to help users to achieve their end-goal of ordering with ease. Unlike its predecessor, the Ronald McDonald bot attempted to hold a conversation with its users, engaging in social niceties and using some markers of engagement. With regards to IBM's Characteristics table, this chatbot would score high on both its primary trait of Agreeableness and secondary trait of Conscientiousness, suggesting that it was helpful, cooperative, polite and considerate. In terms of its results on the Emotion Analyser, the dominant tones in the chatbot text were Joy (0.61) and Tentativeness (0.72), implying that the text gave off a sense of pleasure, safety and contentment but in parts was also questionable or debatable.

While it may seem that a dominant tone of Tentativeness is contrary to the personality in question, it should be noted that the IBM Emotion Analyser is not a perfect indicator of the actual tone of the text and the sheer volume of questions in the text, given the nature of the chatbot, may have resulted in Tentativeness being detected as a dominant tone (See Fig. 1, middle chatbot).

Chatbot C: The third chatbot personality, "Ron", was designed to simulate the behaviour of a bubbly, energetic friend. Unlike the other two chatbots, which attempted to maintain varying degrees of social distance, the intention of the Ron chatbot was to mimic an interaction between close friends, with its tagline reading "Are you ready for the best meal of your life!". To that effect, it made use of several different markers

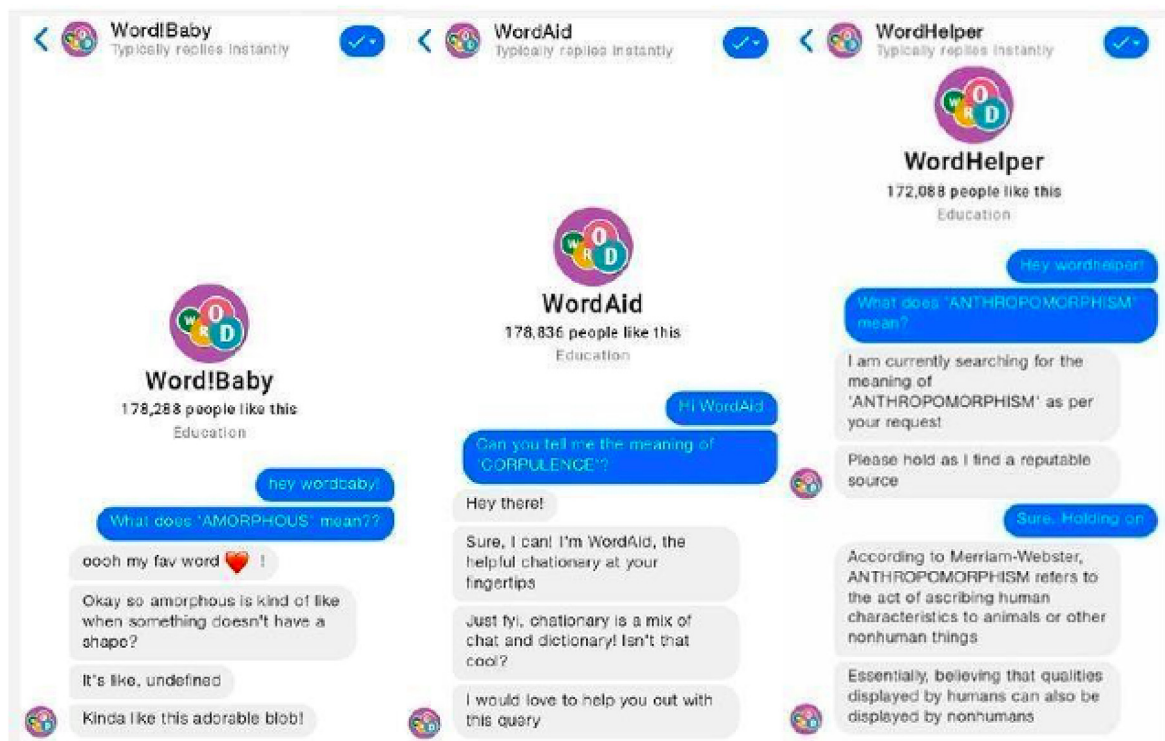


Fig. 2. Friendly, prosocial and transactional chatbot personalities for Experiment II, developed using Botsociety.

of engagement such as emoticons, exclamations marks and capitalized letters. Additionally, it also used references from pop-culture in the form of images, popular slang and acronyms as well as reduplication of letters in order to create the sense of a genuine interchange between friends. Using the IBM Characteristics Table as an indicator, this bot scored highly in both its primary trait of Extraversion and secondary trait of Agreeableness, with the qualitative description of its personality suggesting that it was social, communicative, vibrant and enthusiastic. This was further supported by the results of the text segments fed into the IBM Emotion Analyser, which found that the dominant tones in the Ron chatbot were Joy (0.63) and Confidence (0.66), suggesting that the bot was likely to be perceived as hopeful, assured, pleasing and loving (See Fig. 1, rightmost chatbot).

The skeleton structure of the conversation was the same for all the chatbots – chatbots would introduce themselves to users and ask them to reciprocate, run them through the ordering process, confirm their order and then sign off with a parting message. However, there were individual differences in the chatbot's conversation flows in order to mirror the effects of personality and use cases (see Figs. 1 and 2).

2.3. Chatbot design and pilot testing

The three chatbots: WcBot, Ronald WcDonald and Ron (for further details please see section 2.1) were built using the Bot-Builder Tool, a platform developed by Haptik, over the course of four days. Since the Bot-Builder Tool is a rule-based platform, all the “bot replies”, and “user says” were manually developed by the experimenter and then fed into each of the bots in accordance with their personality rather than being generated by the bot itself. The conversation flows were developed solely by the researcher based on personal perceptions and interpretations of each types of conversation derived from prior literature on the subject (Quan et al., 2018, pp. 4279–4285). It followed the standard structure described in Section 2.1.”. Post bot-building, a 2 h long in-situ focus group was carried out using participants found in Haptik in order to see if potential users would be able to identify the personalities as well as to streamline the chat process so that it mirrored an actual use-case. Nine participants (44% female) from various teams across Haptik, ranging from DevOps to Human Resources (M: 26.3, SD: 1.826) were asked to interact with the three chatbots. Participants spent around 10 min interacting with each chatbot respectively in a within-measures design and then spent the rest of the hour and half pointing out any grammatical issues, problematic statements or things they thought could be improved, with the participants first interacting with the transactional personality (WcBot), followed by the prosocial personality (Ronald WcDonald) and then finally with the friendly personality (Ron). While the main purpose of this focus group was to find and eliminate any bugs in the chatbots, it also proved to be a valuable space for gaining suggestions that could help improve the overall conversation flow of the bots. In order to ensure that the chatbots, which were built solely for the purpose of this study and therefore were only at a demo level, simulated live bots to the best of their ability, participants were audio-recorded as they carried out a Think Out Loud protocol while interacting with each of the three chatbots in turn (Lewis, 1982). In accordance with the rules of the protocol, the participants were usually not interrupted as they vocalized their thoughts but were prompted using very basic questions if they lapsed into silence for over 30 s.

Post data collection, the audio clips were analyzed in order to find the areas in which the bots needed improvement. Based on the responses, three key areas were identified as needing further debugging and fixing — time lags between messages, the aesthetic aspects of the chatbots, such as the website on which the chatbots were displayed or their manner of interacting with the user as well as bot breaks due to unlinked messages or lack of data. These were fixed immediately after in order to ensure a seamless experience for any future participants. With regards to background and UX, the background as well as the logo was standardized for all three chatbot demo pages in order to ensure that they didn't affect

initial impressions and thus lead to biased data. However, since it was important to create some sort of differentiation between the bots, the bots were distinguished by the variations in their names and taglines, as well as website background which were supposed to be indicators of their personality. In addition, upon repeated complaints about the length and redundancy of certain flows, which included the fact that the “*bot asks too many questions*” or “*why is this bot telling me French fries are great, I just want to order*” in the Friendly version of the chatbot, the bot's flow was edited to reduce any unnecessary features such as questions about likes or dislikes or statements about the food. Bot breaks were dealt with by reducing the amount of opportunities in which users could use free text (i.e. users were only allowed to choose between pre-selected reply options rather than inputting their own statements) and constraining them to quick replies, which were also designed in a manner that were indicative of the bot's personalities. Free text, in general, was limited to the Friendly chatbot in order to understand whether it acted as a draw for users and the questions were designed to elicit fairly basic replies. Although users could choose to use free text in the other two bots by typing out the options, such as “*yes, I would like to order!*”, participants usually preferred to use the option buttons since typing out the text seemed to be a fairly redundant exercise.

Apart from the issues with the bots that were pointed out as a result of this exercise, the resultant qualitative data also yielded several new recommendations that evolved into additional features in the bots. Since the bots had been designed to mimic food ordering bots, in this case for the fictional company of WcDonald's, these new features were based on repeated suggestions to increase convenience for users. New features included order confirmation, more quick reply options such as a variety of meal options, ETD estimates and shortcuts on the task box for maximum efficiency, as participants tended to dislike having to start the flow all over again in order to access a different option.

3. Experiment 1

Experiment I tested a food ordering use-case, with the three chatbots being marketed as food- ordering bots for the fictional company of WcDonald's. It was hypothesized that the chatbot with a friendly personality would have higher ratings than the other two chatbot personalities across all three factors being studied, due to the intrinsic appeal of a friendly nature (Smits & Cherhoniak, 1976).

3.1. Methodology

Participants: Forty participants (60% female) aged 18–53 were drawn using convenience sampling (M = 31.25, SD = 12.47). Although there are a number of issues with this particular sampling method, it was used due to the highly technical nature of the study, the somewhat esoteric nature of chatbots as well as the criteria for participation. In terms of the demographic, all the participants were completing or had completed their tertiary education, with 55% of participants holding a Bachelor's Degree and 45% holding a Master's Degree. Ethnically, a large majority of the participants identified as Indian (0.925), while the remaining sample was either Pakistani (0.05) or Russian (0.025). However, all participants were fluent English speakers and lived in urban cities, with access to a smartphone, laptop or both. Previous experience with chatbots was not mandatory as it could have influenced participating ratings due to biased expectations.

Materials: Since the study attempts to study user reactions to different chatbot personalities, there was no need of any specialist apparatus. Participants were required to have access to a single Wi-Fi linked device (preferably either a mobile or laptop) on which they could interact with the three chatbots and then complete a short survey to indicate their sentiment towards various aspects of the different chatbots. The survey in question consisted of twelve questions, out of which three questions were ancillary and collected data about the basic participant demographics of gender, age and level of highest education. The other

nine questions were simply bot-specific reduplications of three core questions which focused on understanding three key aspects of the user's experience with the chatbot; namely, the likelihood of the user choosing to interact with the bot again, their level of comfort with the bot and lastly, their estimation of how productive and efficient their interaction with the bot was. In none of the questions was the bot's personality mentioned; bots were simply referred to as Chatbot 1, 2 and 3. A five-point Likert Type scale was used in order to understand user preferences over these three factors, with one anchoring the negative end and five the positive end of the bipolar scale. Due to the repetitive nature of the questions, three separate Likert type items were preferred over a fully-fledged Likert scale since the several Likert-type questions per factor would reduce the willingness of the participants and may possibly bias data due to fatigue and/or frustration.

Design: The study used a within-measures design, with the chatbot's personality acting as the independent variable. The independent variable was then further divided into three different categories, which were explained in detail under the Chatbot Personality Outline section (section 2.1). The dependent variable in this study was the user's ratings on the Likert type questions for each of the three factors being studied.

Procedure: Participants were first briefly informed about the purpose of the study. Following this, they interacted with each chatbot for a time period of about 2–4 min, and could choose to continue with or switch to the next chatbot at will, as long as they were within the aforementioned time period. Participants were told that they were interacting with a chatbot that could help them in ordering food and were asked to interact with it as if they were actually ordering food, to the best of their ability. The interactions all followed a set order — participants first interacted with the transactional personality (WcBot), followed by the prosocial personality (Ronald McDonald) and lastly, with the friendly personality (Ron). Post interaction, participants were directed to a short Google Form survey that consisted of the twelve questions explained above (See Methodology: Materials).

3.2. Results and discussion

Three separate non-parametric Friedman's tests were conducted in order to understand if any differences existed in the participants' ratings of their likelihood of re-interacting with, comfort level and the productivity and efficiency across the three chatbots. All three tests found that there was a statistically significant difference in ratings for likelihood of reinteraction, $\chi^2(2, N = 40) = 12.86, p < 0.05$; comfort levels, $\chi^2(2, N = 40) = 12.97, p < 0.05$; and productivity and efficiency, $\chi^2(2, N = 40) = 8.13, p < 0.05$. The Kendall's coefficient of concordance for likelihood of reinteraction ($W = 0.161$), comfort levels ($W = 0.162$) and productivity and efficiency ($W = 0.102$) indicated that there were fairly strong differences in user ratings for the three different personalities across the three factors. Following this, a post-hoc analysis for all three factors using Wilcoxon signed ranks with a Bonferroni correction applied was carried out, resulting in a significance level of $p < 0.017$.

Median (IQR) ratings for likelihood of re-interaction across the transactional, prosocial and friendly personalities were 3 (3–4), 3 (3–4) and 4 (3.25–5) respectively. There were no significant differences between the transactional and prosocial trials ($Z = -0.333, p = 0.739, r = 0.05$) or between the prosocial and friendly trials ($Z = -2.352, p = 0.019, r = 0.37$). However, there was a statistically significant difference between the ratings of the transactional vs friendly trials ($Z = -2.560, p = 0.008, r = 0.405$).

For comfort levels, median (IQR) ratings across the three personalities were 3 (3–4) for the transactional personality, 3 (3.25–4) for prosocial and 4 (3.25–5) for friendly. While no significant differences were found for the transactional vs prosocial trials ($Z = -0.079, p = 0.937, r = 0.012$), the prosocial vs friendly trials and the transactional vs friendly trials yielded significant differences ($Z = -3.171, p = 0.002, r = 0.51$; $Z = -3.021, p = 0.003, r = 0.49$), suggesting that the friendly chatbot had much higher ratings for comfort than the prosocial and transactional chatbots.

Lastly, median (IQR) differences for productivity and efficiency were 4 (3–4) for the transactional personality, 4 (3–4) for the prosocial personality and 4 (3–5) for the friendly personality. Significant differences across ratings were found in the prosocial vs friendly trials ($Z = -2.526, p = 0.012, r = 0.41$) and the transactional vs friendly trials ($Z = -2.840, p = 0.005, r = 0.46$), indicating that the friendly chatbots had a higher rating than the transactional and prosocial chatbots. There were no significant differences in the transactional vs prosocial trials ($Z = -0.580, p = 0.562, r = 0.09$).

The results of the study suggest that there was an overall preference for the friendly personality over the other two across the three domains being tested. On the basis of the results of the Wilcoxon Signed Ranks test, it was observed that when it came to pairwise differences, there was a preference for the friendly personality over the transactional one when it came to Likelihood of Re-interaction and a preference for the friendly personality over both transactional and prosocial personalities when it came to Comfort and Productivity & Efficiency. These results (all significant at $p < 0.05$) allow the null hypothesis to be rejected and the alternative hypothesis to be accepted, at least in the context of this study.

4. Experiment II

Experiment II has many conceptual similarities to Experiment I. However, in the case of Experiment II, the transactional personality was called *WordHelper*, and behaved much the same way as *WcBot* from the previous study, giving quick and to the point replies to users; while the prosocial personality for Experiment II was nicknamed *WordAid*, and introduced itself as the 'helpful chationary at your fingertips', attempting to imitate the *Ronald WcDonald* chatbot from Experiment I. Lastly, the friendly personality (*Ron* in the previous use-case) was titled *Word!Baby*, and much like the previous friendly personality, used a variety of engagement markers like emojis, GIFs and slang in order to appear like a friend for the user. The descriptions for each of the personalities remained the same as the ones for Experiment I (See Section 2.1, Fig. 2 for more). However, unlike Experiment I, participants were sent the videos in a randomized order in order to avoid fatigue effects for a particular personality type that may have occurred as a function of participants always interacting with that particular personality at the end.

4.1. Methodology

Participants: 12 participants aged 11–15 (41.7% female) were recruited using convenience sampling for this experiment ($M = 13$; $SD = 1.54$). All participants were currently completing middle school in a metropolitan city – Mumbai (66.67%), Delhi (16.67%), Chennai (8.33%) and Hong Kong (8.33%) and were fluent English speakers. They all identified as ethnically Indian and owned or had immediate access to a smartphone, laptop or both, either had or had access to a Whatsapp account and were aware of the concept of a chatbot. A smaller sample size was tested due to the unavailability of participants fitting the age criteria, as the experiment occurred during the summer holiday period and many potential participants declined to participate.

Design: A within-measures design was used for this experiment as well, with all participants being exposed to the three types of personalities. The sample interactions for all three personalities were designed using the free version of Botsociety, a popular website used for developing prototypes of conversational interfaces with a Facebook Messenger GUI and participant interactions were captured using video recordings that were then shared with the researcher over WhatsApp. Participants were provided minimal instructions and told that the videos were samples of an interaction with an online chat-based dictionary that was being developed as a potential product and that they were participating in a market research study.

Procedure: Participants were sent the videos, which were between 40 s and 1.5 min long, in a random order and after all three videos had been played, participants were asked to choose which of the three

chatbots they would have liked to interact with. Their responses were sent via WhatsApp itself, in order to reduce the rate of drop-out and provide the experimenter with data faster than a call or in-person interaction would. Following this, all interactions with the participants were recorded in a Google document, after which the data was then subjected to a semantic thematic analysis, in order to understand how participants perceived and responded to the three different personalities. The statements were examined for keywords, adjectives or phrases that were indicative of the personality of the chatbot and then grouped together to form an overarching theme of the chatbot's personality as perceived by the user. This provides an insight into the mental thought process of the audience and thus, could help explain the reason why participants were either drawn towards or away from a particular personality.

4.2. Results and discussion

Based on the responses of Experiment II, it was found that even across use-cases and ages, there was a marked preference for a friendly personality in the chatbot. In terms of first preference, most participants stated that they would have preferred to interact with a chatbot like Word!Baby (66.67%), followed by the transactional personality WordHelper (25%) and lastly by WordAid, the prosocial personality (8.33%). Word!Baby proved to be especially popular with the participants, with 8 participants rating it their first choice. They mentioned that while watching the interaction, they often felt like it was just like talking to another human being, using statements like *'it felt like talking to a friend'* or that it was *'pretty cool'*. Similarly, other common sentiments across the participants echoed the fact that Word!Baby was able to deliver the meaning of the word quickly and in a manner that was easy to understand. The addition of gifs, modern-day slang and emojis also served to make the overall experience more enjoyable, with users observing that these qualities would make them far more likely to interact with Word!Baby over the other two personalities, with one participant expressing that *'I think I would interact with Word!Baby because it was easy to talk to and cute, especially since it used fun images and talked like me'*.

However, the three participants who preferred to interact with WordHelper over Word!Baby, stated that they felt that the interaction was much quicker overall, and therefore better for people in a hurry. One participant even observed that in comparison with Word!Baby and WordAid, WordHelper provided a far more detailed explanation of the word it was asked while the other two had a much less elaborate description. In general, there was not much a preference for WordAid, with only one participant stating that they would like to interact with it over the other two because they felt like it provided them with more information than the other two. However, it should be noted that the same participant also mentioned that they would be equally as likely to use WordHelper as they would be to use WordAid, with the choice depending on their mood.

It should be noted, however, that since participants only viewed video clips of the interactions, their preference for the friendly personality may have been influenced by the fact that it was the most interesting video clip, in comparison to the other two. While the randomization of the video clips provides some element of protection against this potential confounding factor, future research studies in this area should strive to keep the interaction between the user and chatbot as realistic as possible in order to yield results with higher ecological validity, such as by deploying these bots on Facebook Messenger and take note of actual interaction through the use of feedback forms or focus groups. Due to the nature of the interface used in this experiment (Botsociety), it was not possible to collect real-time responses, which is something that other studies may prefer to do. However, despite this, the qualitative data is indicative of how the personality of the chatbot could influence its potential users and act either as a draw or deterrent.

5. General discussion

The results of both studies suggest that there is definitely an overall trend that supports the idea that friendly personalities are preferred, even in chatbots. These findings come at a time when research around the intersection of personality and HCI is still at a fairly nascent stage and requires additional support, albeit being leaps and bounds ahead of expectations. The evolution of chatbot personalities is currently enjoying the beginning of its golden age, with incredible findings and developments already being put into practice at the level of social chatbot giants like Microsoft's Xiaoice and several avenues of research dedicated to understanding how personality can be capitalized upon in the domain of HCI. What sets this study apart from other literature on this particular subject is its focus on personality in enterprise chatbots rather than on the development of personality in general conversation chatbots like Mitsuku or the aforementioned Xiaoice, which is largely what existing literature focuses on. Moreover, by using actual chatbots to carry out the study instead of a WOZ protocol, this study provides a far more realistic overview of how users would react to a chatbot personality that was actually constrained by the abilities of the rule-based system rather than to a human attempting to mimic a chatbot.

It also provides additional support not only for the Media Equation Theory by suggesting that humans view interactions with chatbots in the same light as they view interactions with other humans, possibly not only due to the characteristics described by HCI researchers, but also possibly because they equate interactions with chatbots as being similar to interactions with other humans through text via social media platforms (Berry et al., 2000; Reeves & Nass, 1996). Additionally, the study touches upon concepts that bear some resemblance to Mori's *uncanny valley* effect, which postulates a positive correlation between an object's degree of similarity to a human and a human's emotional response to the object. As we become increasingly integrated with social media and begin to use texts and messages as our preferred form of communication, is it possible that chatbots will be able to inspire an *uncanny valley* like response in their human users? (Mori, 1970).

In essence, by carrying out both a quantitative and qualitative analysis of personality preferences, this study provides an insight not only into the general attitude towards chatbot personality but also provides evidence of thought processes that occur while interacting with personable chatbots – something which is vital to improve the quality of future human-chatbot interactions. Although it is difficult to compare quantitative and qualitative data because of their inherent differences, carrying out studies that examine both types of data provides a holistic overview of how users potentially feel about chatbot personalities, and identifies gaps as well as provides multiple avenues of future research that could be explored to better understand personality in chatbots.

However, while this study does provide objective data that helps support the initial hypothesis of people preferring a friendly personality and contribute to the literature available on this subject, it should be noted that there are several areas for improvement in the design and execution of the study that should be taken into consideration for future replications. Arguably, the biggest issue is the structured nature of Experiment I. This leaves it open to ordering effects, especially since participants were either verbally informed of or sent a standard set of instructions with an exact order of interactions (WcBot, Ronald WcDonald and then Ron). The results of the experiment, which supported the idea that friendly personalities are preferable to prosocial or transactional ones may have occurred in part because the participants interacted with the friendly chatbot personality (Ron) at the very end, by which point they may have become extremely comfortable with chatbot usage – which could have acted as a confounding factor. However, since Experiment II yielded similar results after randomization and using the same personalities, it is possible that personality was the key factor and no confounding factors were at play. Although the results from these two

experiments were not comparable given the differences in the study design, leading to some concerns about the internal validity of the study, the similarity in the results and converging conclusions has eased the concern of the possible ordering effects in Experiment I.

Other limitations are focused on participants, who were selected using convenience sampling. This could leave potential room for systemic biases or skewed viewpoints given the relatively similarity of participant demographics, but given that chatbot literature is still at a nascent stage and unfamiliarity with chatbots could have resulted in other biases. However, an additional point to note with regards to participant limitations is that the demographic tested in this paper is the one most likely to initially adapt and use chatbots - urban dwellers who have a fairly decent command of English and own a smartphone or laptop, if not both. Similarly, certain concerns could be raised with regards to the theoretical robustness of using FFM or other deductive models of personality whose purpose is to classify aspects of personality in order to create an entirely new personality from scratch. Additionally, it could also be argued that the rule-based nature of the chatbot personality development could leave it open to the researcher's internal biases of what prototypes of friendly, prosocial or transactional personalities should be like, and that generative models would have provided a more unbiased view. Lastly, one last limitation could be that the examination of the role of personality in HCI was unidirectional and did not take into account the role the user's personality could have on their preferences and perceptions of each of the three types of personalities. Future research in this area is recommended to take into account these limitations in order to provide a more holistic insight into the role of personality in HCI.

6. Future avenues of research

Much of the literature on chatbot personality today focuses on either developing chatbot personality via generative models (See Xiaoice or Mitsuku, for example) or attempting to understand what sort of personality appeals to users the most. However, research that focuses largely on how human personality can impact HCI has focused mostly (with some exceptions) on the sub-area of human-robot interaction (HRI) and not the overarching field of human-computer interaction in general, largely due to the fact that HRI has more parallels with human-to-human interaction (Robert, 2018). Studies in this area have found that people scoring high on extraversion often feel psychologically closer to robots, tend to be far more accepting of closer distances between them and the robot than others and are more likely to humanize robots than others (Gockley & Matarić, 2006; Syrdal et al., 2007). In fact, research suggests that not only are people who score high on extraversion and low on neuroticism more likely to anthropomorphize robots than others, they are also more likely to believe that these robots possess unique characteristics and innately human qualities (Salem et al., 2015). Other studies have found differences with regards to preferences in appearance, with people who score high on neuroticism often preferring mechanical looking robots over more humanoid looking ones (Walters et al., 2008).

Although one could generalize and argue that since HRI is inherently a part of human-computer interaction and therefore any other such interaction (regardless of the medium) should elicit similar responses, it should be noted that interaction with a robot and a chatbot contains fundamental differences. Unlike robots, who often have vaguely humanoid features as well as a physical presence that make the interaction feel more realistic, chatbots can only communicate over text and have a more limited range of expression, often constrained largely to text, emoticons and images — thus rendering this generalization unfair. As a result, an interesting future avenue of research would be to focus specifically on chatbot personality and understand how the personality of the user plays a key role in their perception and continued interaction with various chatbot personalities.

While this particular study hopefully sheds some more light on the growing literature about chatbot personality, it would be a gross mistake to expect or direct all future research in this field to solely focus on developing intrinsic personalities for chatbots. In fact, given the nuances of the relationship between personality, human interaction and HCI in general, it would not be unfair to state that the current literature is only a minute fraction of what could be a gargantuan amount of research endeavors in the making.

7. Conclusion

This study was an attempt to bolster the existing literature on personality in chatbots and how that can affect user preferences and perceptions in the Global South. Despite the issues and limitations that occurred as a result of utilizing actual chatbots instead of a Wizard of Oz protocol or choosing to focus on chatbots that worked on a rule-based system rather than a generative model, this paper found evidence that the presence of a personality in a chatbot, particularly one that is engaging, witty and funny will positively impact users, influencing their perception and usage of the chatbot. Although this study is not without its limitations, it provides support and insight into the importance of personality in human interactions, even when these interactions occur with non-sentient entities.

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Erratum regarding missing Declaration of Competing Interest statements in previously published articles

Declaration of Competing Interest statements were not included in the published version of the following articles that appeared in previous issues of Social Sciences & Humanities Open.

The appropriate Declaration/Competing Interest statements, provided by the Authors, are included below.

1. "The potential role of European professors in the context of Rwandan civil engineering education: Reflections after a teaching experience"(SSAHO, 2020; 2/1: 100019) <https://doi.org/10.1016/j.ssaho.2020.100019>

Declaration of competing interest: The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

2. "Chatbot personality preferences in Global South urban English speakers"(SSAHO, 2021; 3/1: 100131) <https://doi.org/10.1016/j.ssaho.2021.100131>

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3. "Locked up under lockdown: The COVID-19 pandemic and the migrant population"(SSAHO, 2021; 3/1: 100126) <https://doi.org/10.1016/j.ssaho.2021.100126>

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4. "The multifold intertextuality in Lee Chang Dong's burning"(SSAHO, 2021; 3/1: 100119) <https://doi.org/10.1016/j.ssaho.2021.100119>

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5. "Anti-Chinese stigma in the Greater Toronto Area during COVID-19: Aiming the spotlight towards community capacity"(SSAHO,

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6. Branding in social media. Using LinkedIn in personal brand communication: A study on communications/marketing and recruitment/human resources specialists perception"(SSAHO, 2021; 4/1: 100174) <https://doi.org/10.1016/j.ssaho.2021.100174>

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7. "Burnout, self-efficacy and job satisfaction among primary school teachers in Morocco"(SSAHO, 2021; 4/1: 100148) <https://doi.org/10.1016/j.ssaho.2021.100148>

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10. Readings of the coloniality of power in the COVID-19 global dynamics: A brief reflection on Global South's socio-political

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12. The association between polygyny statuses of currently married and in-union women and attitude towards intimate partner violence against women in Ghana”(SSAHO, 2021; 4/1: 100207) <https://doi.org/10.1016/j.ssaho.2021.100207>

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13. The coronavirus trade-off – Life vs. economy: Handling the trade-off rationally and optimally”(SSAHO, 2021; 4/1: 100215) <https://doi.org/10.1016/j.ssaho.2021.100215>

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14. “The possibility to be creative is the reason I want to teach”: Pre-service teachers’ perceptions of creative teaching and philosophical education”(SSAHO, 2021; 4/1: 100190) <https://doi.org/10.1016/j.ssaho.2021.100190>

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15. Towards the hedonic modelling and determinants of real estates price in Morocco” (SSAHO, 2022; 5/1: 100176) <https://doi.org/10.1016/j.ssaho.2021.100176>

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